

Velora — CPRI Hackathon Project Summary

Team: Velora

Project objective

Build a reproducible measurement-audit and prediction pipeline that evaluates sensor measurements in their operating context, classifies test validity, and predicts the hidden Reference_Parameter for the CPRI screening dataset.

Dataset

The official participant dataset contains 1,000 training records and 350 test records. The training data contains 11 columns including Reference_Parameter and Validity_Label; the test data contains 9 input columns. The final ground-truth verification matched the repository data to the official participant PDF with zero cell mismatches in both training and test datasets.

Data quality and preprocessing

The audit explicitly preserves suspicious duplicate observations rather than deleting them. Missing sensor values are handled reproducibly: S1–S3 use leakage-safe Ridge-based imputation, while S4 remains unimputed with a missingness indicator. Sentinel and other anomalous measurements are retained as evidence for the validity audit.

Operating-regime analysis

KMeans operating partitions were evaluated and $k=4$ was selected for balance and interpretability. These clusters are treated as empirical operating partitions rather than being claimed as proven physical regimes.

Cross-sensor consistency

A Ridge model with $\alpha=1.0$ provides the authoritative cross-sensor consistency engine. OOF performance: MAE 0.494530, RMSE 1.672792, R^2 0.893817.

Validity classification

The frozen Random Forest validity model uses an operational threshold of 0.236 selected using development folds, with the 200-row holdout kept isolated. Primary locked-holdout results at threshold 0.236: TN=170, FP=3, FN=3, TP=24; Invalid precision=0.888889, recall=0.888889, F1=0.888889.

Reference Parameter regression

The frozen GradientBoostingRegressor uses 100 estimators, $\text{max_depth}=3$, $\text{learning_rate}=0.1$, and $\text{random_state}=42$. Locked holdout: MAE 1.257678, RMSE 3.613879, R^2 0.887163, Median AE 0.435748, bias +0.038177. Valid subgroup MAE=0.750612 and Invalid subgroup MAE=4.506471.

Audit / attention scoring

The frozen attention score combines negative/sentinel flags, cross-sensor inconsistency, S3 residual extremeness, imputation and missingness counts, validity-model uncertainty, and operating-support. It is an audit heuristic, not a formal uncertainty interval or universal out-of-distribution detector.

Leakage prevention and reproducibility

Preprocessing and model fitting are kept within the appropriate validation folds. The locked holdout is excluded from threshold optimization. Random state is 42. The final verification suite reports 291 passing tests and zero failures.

Final submission

The verified prediction file contains exactly 350 rows and three columns: Test_ID, Validity_Label, Reference_Parameter. The final Velora.csv is a byte-for-byte copy of the previously verified submission, with SHA-256 5586561015bec10b02ebf00a2491e42b9670ab6b8ce54aed89e0393e3dfaf67c.

Production status

CP12.1 PASS — ACTUAL CPRI DATASET VERIFIED. Production artifacts are frozen and the submission is verified. No further model changes are authorized.